SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 2 (TOPIC 2)

Theme: MICROCOMPUTER COMPONENTS, SUBSYSTEMS & ARCHITECTURAL COMPARISONS

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 2.1 (Devices & Logical Configurations: CPU, Memory, Bus, I/O, Storage), Subtopic 2.2 (Comparative Aspects)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 4 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Support your explanations with clear examples, labeled diagrams, or command syntax where requested.
4. Use spreadsheet formulas only where requested in the spreadsheet assignment; no other numerical calculations are required.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL ARCHITECTURAL CONCEPTS [25 MARKS]

Question 1 [5 Marks]: Detail the functional role of the four internal CPU components: (a) Arithmetic Logic Unit (ALU), (b) Control Unit (CU), (c) Program Counter (PC), and (d) Instruction Register (IR). Describe how they collaborate during instruction decoding.

Question 2 [5 Marks]: Differentiate between Static RAM (SRAM) and Dynamic RAM (DRAM) at the transistor/circuit level. Explain why SRAM is utilized exclusively for processor cache hierarchies (L1, L2, L3) while DRAM is utilized for main system memory.

Question 3 [5 Marks]: Explain the mechanical architecture of a Hard Disk Drive (HDD). Define the components that govern total average data access time: Seek Time, Rotational Latency, and Transfer Time. How does a Solid-State Drive (SSD) eliminate two of these components?

Question 4 [5 Marks]: Define the three functional sub-buses comprising the system bus: (a) Data Bus, (b) Address Bus, and (c) Control Bus. Specify which of these buses are unidirectional and which are bidirectional, providing rigorous architectural reasons.

Question 5 [5 Marks]: Compare CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer) design philosophies. Explain how modern microprocessors (such as Intel Core x86-64 and Apple M-Series ARM) incorporate elements of both philosophies.

SECTION B: HARDWARE CONCEPTS & COMPARISONS [35 MARKS]

Question 6 [10 Marks]: Describe the fetch, decode, and execute stages of the processor instruction cycle. Explain how the control unit, arithmetic logic unit, and registers contribute. [10 Marks]

Question 7 [10 Marks]: Compare RAM, ROM, cache, and secondary storage by purpose, volatility, and typical use. Give one example of when each is relevant to a student or researcher. [10 Marks]

Question 8 [8 Marks]: Explain the difference between a hard disk drive and a solid-state drive. Discuss durability, moving parts, and suitable uses without performing capacity or speed calculations. [8 Marks]

Question 9 [7 Marks]: Describe the purpose of the address, data, and control buses. Explain how an input device and an output device communicate with a computer system. [7 Marks]

SECTION C: SYSTEM COMPARISONS & APPLIED INTEGRATION [40 MARKS]

Question 10 [15 Marks]: Workstations versus Standard Commercial Microcomputers:
The Department of Soil and Geological Sciences at SUA requires a high-performance computer for running 3D digital elevation models (DEM) and hyperspectral soil GIS simulations.
(a) Contrast an engineering workstation with an ordinary consumer desktop PC across: Processor architecture (e.g., AMD Threadripper/Intel Xeon vs Core i5), Memory architecture (ECC vs non-ECC), GPU capabilities (Nvidia RTX Ada Generation vs consumer GeForce), and Storage interfaces. [7 Marks]
(b) What is Error-Correcting Code (ECC) memory? Explain the physical Hamming code mechanism that enables single-bit error correction and double-bit error detection (SEC-DED). Why is this critical for long-running scientific simulations? [5 Marks]
(c) Recommend a workstation or a standard desktop for this research task. Justify your choice using software needs, reliability, support, and expected use. [3 Marks]

Question 11 [15 Marks]: Comparative Analysis: Microcomputers vs Mainframes and Minicomputers:
(a) Construct an exhaustive comparison matrix evaluating Microcomputers, Minicomputers (Midrange systems), and Mainframes across: Typical user concurrency, I/O processor subsystem architecture, operating system complexity, physical footprints, and fault-tolerant hardware redundancy. [8 Marks]

(b) Trace the historical paradigm shift that led to the decline of traditional minicomputers (e.g., DEC PDP and VAX) in favor of networked microcomputer client-server architectures in university and enterprise environments. [4 Marks]
(c) Explain the concept of 'Symmetric Multiprocessing' (SMP) and how modern multi-socket microcomputer servers achieve processing capabilities that rival historic enterprise mainframes. [3 Marks]

Question 12 [10 Marks]: Peripheral Interface & Agricultural Sensor Integration:
An engineering student at SUA is designing an automated greenhouse control unit based on an embedded microcomputer (Raspberry Pi 4B) connected to soil moisture sensors, temperature probes, and irrigation solenoid relays.
(a) Differentiate between Analog-to-Digital Conversion (ADC) and Digital I/O (GPIO). Why does the microcomputer require an external ADC (such as the ADS1115) to read analog resistive soil sensors? [4 Marks]
(b) Contrast the I2C, SPI, and UART serial communication protocols in terms of wiring complexity, general communication speed, and multi-device addressing. Recommend a suitable protocol for connecting identical environmental sensors. [6 Marks]